Rohan Batra

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SUMMARY

Final-year B.Tech Computer Science (Cyber Security) student at UPES (graduating 2027, CGPA 8.1), open to any role at HPE — networking, platform, software, support or security, internship or full-time. Co-built and operates a LAN-first campus emergency communication system on HPE Juniper SRX, EX and Mist hardware, presented to Team HPE in Bengaluru, and troubleshoots it to root cause. Built Shimpaku, a Mist-style assurance system for a seven-device Junos lab with a guardrailed AI agent. Programs in Python and C++; comfortable across Linux, containers, Kubernetes, cloud and the networking beneath them.

EDUCATION

UPES Dehradun — B.Tech (Hons.) Computer Science Engineering, Cyber Security *Aug 2023 – Aug 2027*
CGPA 8.1/10

SKILLS

Languages: Python, C++, C, Rust, TypeScript, JavaScript, Shell
Networking: HPE Juniper SRX / EX / Mist, Junos, OSPF, BGP EVPN-VXLAN, BFD, gNMI, Virtual Chassis, VLANs, QoS / CoS, LLDP-MED, PoE, NETCONF, DHCP / NAT, DNS, Subnetting, pfSense
Infrastructure: Linux, Kubernetes, Docker, Helm, Nginx, Proxmox, QEMU, Terraform / OpenTofu
DevOps: GitHub Actions, CI gating (Ruff, ShellCheck, gitleaks), cargo-llvm-cov, cargo-mutants
Security: Gitleaks, HashiCorp Vault, Semgrep, Trivy, age encryption, AppArmor, Burp Suite, Cerbos
Data: MongoDB, Redis, PostgreSQL, ChromaDB, MySQL, DuckDB, DuckLake
Backend: FastAPI, Flask, Pydantic, SQLAlchemy, Alembic, Microservices, REST APIs

PROJECTS

UPES-ECS — Campus Emergency Communication System
Live demo: upes-ecs.rohanbatra.in · Code: github.com/rohanbatrain/UPES-ECS
LAN-only emergency calling on HPE Juniper networking. Presented to Team HPE at HPE's Bengaluru campus (15 Jul 2026) and demonstrated live at UPES (20 Jul 2026).
Juniper SRX/EX/Mist, Junos, Asterisk/PJSIP, FastAPI, NETCONF, PowerShell

- Co-built a **LAN-only** campus emergency calling system (one-number 111 hotline) on Asterisk/PJSIP requiring no internet, SIM, or cloud — designed, deployed and operated end-to-end on an **HPE Juniper** networking backbone.
- Configured the enterprise fabric: **Juniper SRX300/320** as the guarded edge (firewall, DHCP, NTP, LAN-only boundary), a **Virtual Chassis of two EX2300-C-12P** PoE switches carrying the voice network, and two **Mist AP32** Wi-Fi 6 access points for campus-wide softphone reach.
- Engineered the voice fabric for zero-touch operation and self-healing: **LLDP-MED** auto-provisioning of voice VLAN, PoE class and DSCP; strict-priority **CoS for RTP**; **DHCP-snooping with Dynamic ARP Inspection** to protect the failover path; and Junos **event-options** policies that power-cycle a hung AP without human intervention. Budgeted PoE against the switch's real 124 W ceiling.
- Designed **offline caller location by switch port** (IP to MAC to port to room) so the network identifies where a 111 call originated with no cloud or location service, and fed live switch/AP state to the Operations Console over **NETCONF**.
- Operated it as a production system where downtime is a safety defect — active/standby high availability, health checks, load testing, and incident root-cause analysis. Three documented RCAs: a serialized single-threaded proxy, a NAT media-address fault dropping every call at 32 seconds, and leaked channels with no teardown path.
- Developed a FastAPI status/control API exposing real-time queue, call, and system state to an operations console and live LED-TV wallboards, provisioned end-to-end via a single PowerShell installer.
- Implemented multilingual IVR across **44 languages** with Piper TTS voice prompts and an offline panic-coach fallback, gated by CI (Ruff, ShellCheck, PSScriptAnalyzer, gitleaks).

Shimpaku — Mist-style Assurance for a Junos Lab *2026 – Present*
Code: github.com/rohanbatrain/shimpaku · Lab: github.com/rohanbatrain/juniper-7-day-sprint
Assurance NMS for a seven-device vJunos lab, built on my Juniper 7-Day Sprint lab as code.
Python, FastAPI, React, TypeScript, Junos, gNMI, NETCONF/PyEZ, MCP, DuckDB, SQLite, OSPF, EVPN-VXLAN, BFD, Ansible, libvirt/KVM

- Built **Mist-style assurance**: gNMI telemetry every 10 s, service levels from 24 synthetic probes, and **blast-radius** root cause on live devices, with a fault-injection harness (80/80 self-test trials over 16 scenarios, config restored after each).
- Designed **Ask Shim**, an MCP AI agent that investigates across campus and leaf–spine topologies but can only **propose** changes: human approval, ten guardrails, `commit confirmed` with a validation window, and a hash-chained audit of every call — in its first week, 8 of 16 proposals were blocked before any commit.
- *Juniper 7-Day Sprint (the lab underneath):* **topology-as-code** vJunos on plain libvirt/KVM — per-link bridges, nftables-isolated management, qcow2 snapshots for instant rollback after every "break it" drill.

- Built a spine/leaf **EVPN-VXLAN** fabric (OSPF underlay, iBGP EVPN overlay, ECMP); a dead spine took the 40 s OSPF dead interval to detect, so I closed it with **BFD** (300 ms × 3) as an 8-line generated diff.
- Automated it end to end: config rendered from the topology, committed over **NETCONF (PyEZ)** or Ansible, then OSPF, ECMP, EVPN and cross-VXLAN reachability asserted — idempotent, drift-repairing, checked in GitHub Actions CI.

Paridyum — Independent Multi-Tenant Platform
Self-directed systems project: built solo from an empty directory to a running five-node Kubernetes cluster on self-hosted metal with a public internet edge.
Rust, OpenTofu, Ansible, Talos Linux, Kubernetes, Cilium, PostgreSQL

- Built a **specification-as-software governance system**: 251 versioned artifacts (constitution, architecture decision records, standards, RFCs) with machine checks mapped outward to **NIST SSDF, CIS v8.1 and NIST CSF 2.0**, a multi-axis test-type taxonomy, and a precedence order in which existing code is the lowest authority.
- Delivered the full infrastructure path: hardened bare metal (AppArmor, nftables, confined QEMU/libvirt) through **Proxmox and OpenTofu** to **Talos Linux and Kubernetes** with Cilium as CNI and kube-proxy replacement, L2-announced load-balancer VIPs, and etcd snapshots that assert they captured something.
- Wrote **31,000+ lines of Rust** across two production tools — hexagonal architecture, trait-based abstraction, injectable clock adapters, and three independent SHA-256 hash chains for tamper-evident audit records.

AutoTriage — Autonomous Vulnerability Triage & Remediation Agent
Code: github.com/rohanbatrain/autotriage
An agent that triages real scanner findings end-to-end, with evals and human-in-the-loop guardrails.
Python, Claude Agent SDK, Semgrep, Trivy, Gitleaks, Pydantic

- Built an autonomous vulnerability-triage agent on the Claude Agent SDK that ingests Semgrep, Trivy, and Gitleaks findings, reasons about severity and business impact, and files tickets, assigns owners, and drafts remediation PRs.
- Enforced human-in-the-loop guardrails via Pydantic validators — confidence-threshold escalation, fail-closed on malformed output, and prompt-injection defenses that treat scanner output as untrusted input.
- Built a fix-validation loop that applies proposed patches to an isolated copy of the code and re-runs the scanner, accepting a fix only when the finding is verifiably resolved with no regression.
- Designed an eval harness scoring triage decisions against a labelled ground-truth set: **100% verdict accuracy** across 44 live findings triaged into 24 tickets and 20 human escalations, backed by 68 tests at 89% coverage.

EXPERIENCE

Hardware & WireGuard Research Intern *Jul 2026 – Present*
Exaflair Technologies Pvt. Ltd., Dehradun — Remote

- Primary infrastructure contributor to **DEKLO**, Exaflair's self-hosted edge PaaS: Ansible-automated **k3s** fleet over a **WireGuard** mesh — a Mumbai hub plus edge nodes on consumer broadband in Chandigarh — with node hardening, fleet conventions, and CloudNativePG PostgreSQL.
- Built backups for every tenant database with restore drills, turned on container-registry authentication, and surfaced node-install failures and live node state in the provisioning portal.
- Wrote the product-research site behind the platform design: instrumented trials, recorded decisions, and measured cross-carrier latency and packet loss.

Platform Engineering Intern *Nov 2025 – Jul 2026*
Eve (TechyMonks), Bengaluru — Remote

- Founded the **Eve Gateway Stack** on APISIX and etcd — central routing, JWT authentication, rate limiting, CORS, and Prometheus metrics — establishing the platform's central trust and routing layer, with gateway-level tenant isolation, trusted tenant/user header injection, and Cerbos-based authorization rules.
- Founded and led the **Eve Data Management Service**: a FastAPI microservice with source/destination/connection APIs, SQLModel models, Alembic migrations, an Airbyte Config API client, sync orchestration, webhook routes, health checks, and Kubernetes/EKS deployment.
- Built **Airbyte-based ingestion** — typed connectors, connection testing, sync polling and S3/MinIO bucket creation — and resolved staging failures across timeouts, secrets, readiness probes and Temporal.

CERTIFICATIONS

- **Open Learning — Migrating from the Cisco CCNA to the JNCIA-Junos**, HPE Networking Education Services (Sep 2026)
- **In progress, via the Juniper 7-Day Sprint:** JNCIA-Junos, JNCIA-Cloud, JNCIA-SEC, JNCIA-MistAI, JNCIA-Design, JNCIA-DevOps, JNCIA-DC

ACHIEVEMENTS

- **Finalist** — HACKIITK 2026, IIT Kanpur: top 0.4% of 1,300+ teams (9,000+ participants), with an offline AI policy-gap analyser (Code: github.com/rohanbatrain/hackiitk)
- **Top 15%** — TryHackMe cybersecurity learning platform
- **Semifinalist, twice** — Eureka! Junior, E-Cell IIT Bombay (2021, 2022)